

Alec T Chen

thealec.chen@mail.utoronto.ca | [Portfolio](#) | [linkedin.com/in/thealecchen](https://www.linkedin.com/in/thealecchen)

EDUCATION

University of Toronto

Sep 2023 – present

Bachelor of Applied Science in Mechanical Engineering – CGPA 3.97/4.00

Toronto, ON

- Received: University of Toronto Scholars (\$10,000), Joyce A. Memorial (\$1,350) and BC Provincial (\$1,250) scholarships.
- Relevant Courses: Manufacturing Engineering, Circuits (LTspice), Solid Mechanics, Computer Fundamentals (C)

RELEVANT EXPERIENCE

Researcher/Robotics Engineer

May 2025 - Present

Laboratory for Extreme Mechanics and Additive Manufacturing – University of Toronto

Toronto, ON

- Leading the design and construction of a self-driving lab containing a 6-axis robotic arm and directed energy deposition to achieve autonomous high-throughput alloy discovery.
- Researched, synthesized, and categorized **105+ publications** to **co-author** a comprehensive review paper on the multi-functionalities of **additive manufacturing** and directed energy deposition (DED).
- Designed, modeled and 3D printed an adjustable fixture using **SolidWorks** to mount and adjust the focal distance of a \$27,000 IR camera to the lab's DED machine.
- Machined **64+ samples** with precision using wire EDM, mill, or drill press and polished them down to **0.05 μm** .

Mechanical Lead

Oct 2024 – Present

Spark! Design Team – University of Toronto

Toronto, ON

- Leading 2 students to design, prototype, and assemble a 600 mm wide spinning arena with a gearbox and **24+** custom-designed components using **SolidWorks**.
- Engineered and fabricated a compact, LED-embedded chessboard featuring **165+** custom **3D-printed parts**, optimizing thickness for embedded magnets while minimizing deflection.
- Designed and **3D printed** 4 unique chess pieces using Blender and **SolidWorks** while adhering to a low-poly aesthetic.

Mechanical/Electrical Team Member

Sep 2023 – Jan 2025

University of Toronto Super Mileage – University of Toronto

Toronto, ON

- Collaborated with 2 teammates in a **high-stress, high-stakes** environment during the **2024 Shell Eco-Marathon**, to bleed our car's brakes by reassembling the entire steering assembly within **one hour**.
- Constructed a **2m** long car stand and workstation with **5 drawers** out of plywood sheets using jigsaws and hand drills.
- Helped develop the team's first-ever **motor controller** by **soldering** electronic components using custom **Altium** schematics and troubleshooting with **oscilloscopes**.

PERSONAL PROJECTS

Custom 3x3 Keyboard

July 2025 – Present

- Designed, modeled, and manufactured a custom keyboard containing 9 keys, a rotary encoder and an OLED screen in **SolidWorks**.
- Designed and ordered a **custom PCB** for the switches using **KiCAD**.
- Developed **custom firmware in C** using QMK to create **39+** unique functions to optimize my **SolidWorks** workflow.

Custom Digital Photo Frame

April 2022 – August 2022

- Designed, programmed, and assembled a **smartphone-controllable** photo frame that can double as a monitor.
- Manufactured an enclosure using MDF boards to integrate electronic components, including a Raspberry Pi, cooling fan, 11" monitor, HDMI switcher, and monitor control board.
- Adapted and debugged outdated **Python scripts** from **GitHub** to achieve custom photo imports and smartphone controllability on the **Raspberry Pi**.

TECHNICAL SKILLS

Design Software

SolidWorks, Fusion 360, Blender, KiCAD, LT Spice, Altium

Machining and Power Tools

Laser DED, Mill, Drill Press, Jigsaw, Hand Drill

Programming

Python, C, MATLAB, LaTeX